LAB 6 – ATOMIC SPECTRA

OBJECTIVES

- Use diffraction gratings for spectroscopy.
- Identify gases based on their emission spectra.
- Characterize light sources based on their spectra.

LAB OVERVIEW

In this lab we continue our exploration of modern physics by studying various spectra. Our primary concern will be observing the emission spectra of various elements, including a verification of the Rydberg formula for hydrogen. We will also qualitatively study some other spectra.

EQUIPMENT

• Optical bench with bench stands	• Spectrometer Scale
• Diffraction Grating	• Discharge Tube Apparatus
• Gas discharge tubes: H, He, *B*, and *A* or *C*	• Lamp with incandescent bulb
• Lamp with CFL bulb	• Slide Holder
• Screen	• Plastic Protractor
• Grating Mirror	

WHAT TO TURN IN

For Experiments 1 and 5 you should submit your answers as you would a normal homework assignment or informal lab. You will submit a lab report for Experiments 2 and 3.

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THEORY AND BACKGROUND

Introduction

In this lab we will continue our exploration of early quantum mechanics by studying the emission spectra of a few different elements.

Spectroscopy

Spectroscopy refers to the study of the light emitted or absorbed by matter, usually taking the form of classifying the emitted or absorbed intensities as a function of wavelength or frequency. Spectroscopy was integral to the development of quantum mechanics. For example, study of the ***blackbody spectrum*** showed that the intensity of light emitted in ultraviolet frequencies falls to zero rather than increasing (as was predicted by the wave picture of light). Another important set of spectra are ***atomic emission spectra***. It was observed that a hot gas under low pressure emitted light primarily in a certain discrete set of wavelengths. This was called a ***bright-line spectrum*** since when the light was passed through a prism spectrometer or a diffraction grating, the light separated into a discrete set of clear, bright ***spectral lines*** - each line representing a wavelength. This is caused by the quantization of energy of electrons in atoms - the electrons can only take on certain discrete energies. When an electron falls from one energy to a lower energy, it must emit exactly the right amount of energy in the form of a photon.[1] Since the energies are discrete, so are the emitted or absorbed photon energies, resulting in the observed spectra. For the purposes of today's lab, we will be concerned with the visible parts of the bright-line spectra, since these will be easiest to directly observe.

Hydrogen

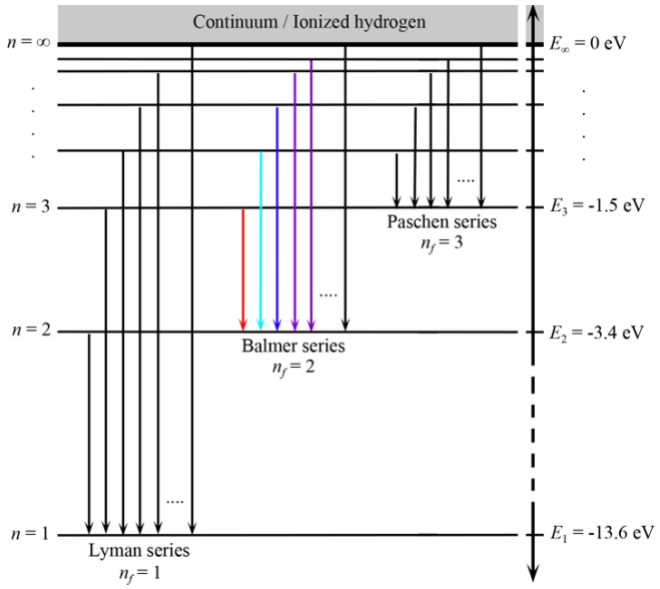


Figure 1: The energy levels and emission spectrum of hydrogen.

Physicists observed emission spectra far before the discovery of the key ideas of quantum mechanics. While they didn't yet understand the *mechanism* behind the particular atomic spectral lines, they were able to come up with some empirical rules, particularly for hydrogen. It was observed that the wavelengths of the emission spectra of hydrogen were given by the ***Balmer-Rydberg formula***,

$$\frac{1}{\lambda} = R\left(\frac{1}{m^2} - \frac{1}{n^2}\right), \tag{1}$$

[1] Actually, there *is* some small wiggle room in the exact value of the energy emitted due to the energy-time uncertainty principle and the decay rate of the energy state in question. This shows up as an additional line width in the atomic spectrum.

where m and n are positive integers with $m > n$ and R is the ***Rydberg constant***,[2]

$$R = 1.0973731568508(65) \times 10^7 \text{ m}^{-1} \approx 1.10 \times 10^7 \text{ m}^{-1}. \quad (2)$$

Johannes Rydberg presented a version of this formula in 1888. The wavelengths are divided into groups based on the value m:

- $m = 1$: The ***Lyman series***, observed by Theodore Lyman in 1906. These are all ultraviolet wavelengths.
- $m = 2$: The ***Balmer series***, observed by Johann Balmer in 1885. These are all visible wavelengths.
- $m = 3$: The ***Paschen series***, observed by Friedrich Paschen in 1908. These are all wavelengths.
- Higher n are all infrared wavelengths. The ***Brackett, Pfund,*** and ***Humphreys series*** (n = 4, 5, 6) were observed in 1922, 1924, and 1953, respectively.

The Bohr Model of Hydrogen

The Bohr model is essentially a 'planetary model' of the atom with a quantization assumption added in. The atom is divided into two parts: The ***nucleus***, a small ball of positive charge $+Ze$ at the center of the atom, and an ***electron*** orbiting the nucleus. Z is the ***atomic number*** and determines the element under consideration ($Z = 1$ for hydrogen, for example). The nucleus is, at a minimum, 1800 times as massive as the electron, so we will treat the nucleus as a point charge fixed in place at the center of our system (the 'sun' of our planetary model).

In the Bohr model, the electron (mass m_e) moves with speed v in a circular orbit of radius r about the nucleus. In 1913, Bohr proposed a quantization rule for angular momentum,[3]

$$L = nh/2\pi. \quad (3)$$

A direct consequence of this rule is that the possible energies of the electron in the atom is quantized according to

$$E_n = -\frac{Z^2 E_0}{n^2}, \quad E_0 = \frac{m_e k^2 e^4}{2h^2} = 13.6 \text{ eV}. \quad (4)$$

The constant E_0 defines a unit of energy called the ***Rydberg***, 1 Ry ≡ 13.6 eV. The integer n refers to the ***energy level*** of the electron.

Normally, the sole electron in hydrogen is in the lowest available energy state, $n = 1$, called the ***ground state***. The energy required to free the electron from the hydrogen (and therefore *ionize* it) is the energy required to increase the energy from the ground state up to an energy of zero. Thus, the ***ionization energy*** of hydrogen is predicted to be equal to the Rydberg, 13.6 eV, by the Bohr model.

A hydrogen atom may also be ***excited*** to a higher state (n = 2, 3, 4, …), where the electron is still bound to the nucleus but has a higher energy. Electrons can be knocked into higher energy states by collisions or the absorption of a photon. In hydrogen, the electron will live in the excited state for a very short time (typically on the order of 10^{-8} s) and then fall down to a lower available state. This can be either another lower excited state or the ground state. By conservation of energy, this de-excitation necessitates the release of a photon whose energy is equal to the difference in energy between the two states of the electron. Since the electron energies are quantized, so must be the emitted wavelengths. Using Eq. 4, it is found that the possible emitted wavelengths are

$$\frac{1}{\lambda} = \frac{E_0}{hc}\left(\frac{1}{n_i^2} - \frac{1}{n_f^2}\right), \quad (5)$$

This is precisely the form of the empirical formula Eq. 1 found by Rydberg. As a bonus, it proves a derivation of the Rydberg constant from the more fundamental constants h (Planck's constant), c (the speed of light), k (Coulomb's constant), e (the fundamental charge), and m_e (the mass of the electron). It also provides a natural interpretation for the integers m and n - these are the initial and final states of the electron as it de-excites, respectively.

[2] No, you don't need to know the value of R to nearly this accuracy. However, I included it because, according to Wikipedia, it is (as of 2012) one of the most accurately measured fundamental physical constants!

[3] Niels Bohr (1913). *"On the Constitution of Atoms and Molecules, Part I"*. *Philosophical Magazine*. **26** (151): 1–24.

It is amazing that Bohr's addition of a single quantum concept (the quantization of angular momentum) to a mostly classical model so nicely predicts the spectrum of hydrogen. It's even more amazing the Bohr didn't provide much of a motivation or justification of his quantization rule beyond the fact that, well, it *worked*! However, the model is not without its failures. Most importantly, it doesn't really work for multi-electron atoms (such as neutral helium). It also fails to account for the ***fine structure*** of hydrogen - small deviations from the predicted Bohr energies due to relativistic effects and the spin of the electron. It doesn't explain the ***Zeeman effect*** - the observed splitting of spectral lines due to the presence of an external electric field. It doesn't account for the relative intensities of spectral lines (as you saw with the mercury spectrum, some spectral lines are much brighter than others). To properly explain these effects, we need the full force of quantum mechanics.

The Spectrum of Hydrogen

The quantization of energy of an electron in hydrogen is responsible for the spectrum of hydrogen. In particular, when the electron is ***bound*** to hydrogen its energy can only take on discrete or quantized values. If the electron has enough energy, it can break free from the hydrogen, creating an ***ionized electron***. The energy of an ionized electron can take on a continuum of values. This is shown as the gray region above the $E = 0$ level in Figure 1.

The spectrum of hydrogen is in essence its thumbprint. The particular set of emitted wavelengths is a unique signature of hydrogen. Here are some features of the hydrogen spectrum based on Eqs. 1 and 6, as well as the relationship of the energy and wavelength of a photon we explored in the previous lab, $\lambda = hc/E_\gamma$.

n_f	n_i	E_γ (eV)	λ (nm)
1	2	10.2	122
	∞	13.6	91.2
2	3	1.89	656
	4	2.55	486
	5	2.86	434
	6	3.02	411
	7	3.12	397
	∞	3.40	365
3	4	0.661	1880
	∞	1.51	821

Table 1. The transitions and spectral lines of hydrogen.

Lyman series ($n_f = 1$):

The ***Lyman break*** is the largest-energy transition from a bound state to $n_f = 1$. There are a continuum of energies above the Lyman break due to a free electron becoming bound. From the table above, the Lyman break occurs at $\lambda = 91.2$ nm.

Note that all emitted wavelengths in the Lyman series are in the ultraviolet part of the electromagnetic spectrum.

Balmer series ($n_f = 2$):

The ***Balmer break*** is the largest-energy transition from a bound state to $n_f = 2$. There are a continuum of energies above the Balmer break due to a free electron becoming bound. From the table above, the Balmer break occurs at $\lambda = 365$ nm.

Lyman-break transition: $n_i = \infty$. Photon energy $E_\gamma = 3.4$ eV. Wavelength $\lambda_{\infty \to 2} = 365$ nm.

Lowest-energy transition: $n_i = 3$. Photon energy $E_\gamma = 1.89$ eV. Wavelength $\lambda_{2 \to 1} = 656$ nm.

Paschen series ($n_f = 3$):

The ***Paschen break*** is the largest-energy transition from a bound state to $n_f = 3$. There are a continuum of energies above the Paschen break due to a free electron becoming bound. From the table above, the Paschen break occurs at λ = 821 nm. Note that this is in the infrared and thus the continuum beyond the Paschen break will show up in visible wavelengths.

Note that all emitted wavelengths in the Paschen series are in the infrared part of the electromagnetic spectrum.

Other Atomic Spectra

The Spectrum of Helium

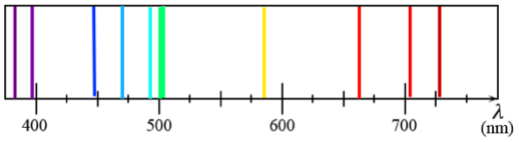
The interaction between the two electrons in helium cause some interesting things happen to the spectrum. The spectrum "splits into two" on the basis of how the electron spins compare, with ***parahelium*** having symmetric (roughly "parallel") spins and ***orthohelium*** having antisymmetric (roughly "antiparallel") spins. Based on various quirks of quantum mechanics, the dominant transitions only occur between two parahelium or two orthohelium states. As in hydrogen, the states are labeled by a principal quantum number n and an angular momentum quantum number l, with the energy depending on both n and l. However, the energy levels no longer follow the Bohr model. Table 2 shows the low-energy states of helium.

State Label	Parahelium	Orthohelium
1s	-24.587 eV	--
2s	-3.972 eV	-4.768 eV
2p	-3.369 eV	-3.623 eV
3s	-1.667 eV	-1.869 eV
3p	-1.500 eV	-1.580 eV
3d	-1.513 eV	-1.514 eV
4s	-0.914 eV	-0.993 eV
4p	-0.845 eV	-0.880 eV
4d, 4f	-0.85 eV	-0.85 eV

Table 2. Energy levels of helium. From http://www.doylegroup.harvard.edu/wiki/images/c/c1/Sansonetti_2004_atomic_spectro_data.pdf

The ***allowed transitions*** (meaning that they have a relatively high probability of occurring compared to other transitions) between these states is shown in Figure 2 and the allowed transitions that emit photons in the visible spectrum (corresponding to energy differences between roughly 1.65 eV and 3.10 eV) is listed in Table 3. Note that parahelium states only have allowed transitions with other parahelium states and orthohelium states only have allowed transitions with orthohelium states.

The locations (but not intensities) are plotted on the spectrum shown to the right. This is our theoretical prediction for observations of the helium spectrum. Note that some wavelengths that are close together may appear as a single spectral line. Some of the spectral lines may be too faint to see.



As a neat historical note, helium was *discovered* due to its emission spectrum! In 1868, Pierre Janssen noticed a spectral line at 587 nm in the spectrum of the sun while observing a solar eclipse. This line didn't correspond to any known emission spectra at the time so it was hypothesized that it came from a new element, which was dubbed *helium* after the sun (helios-).

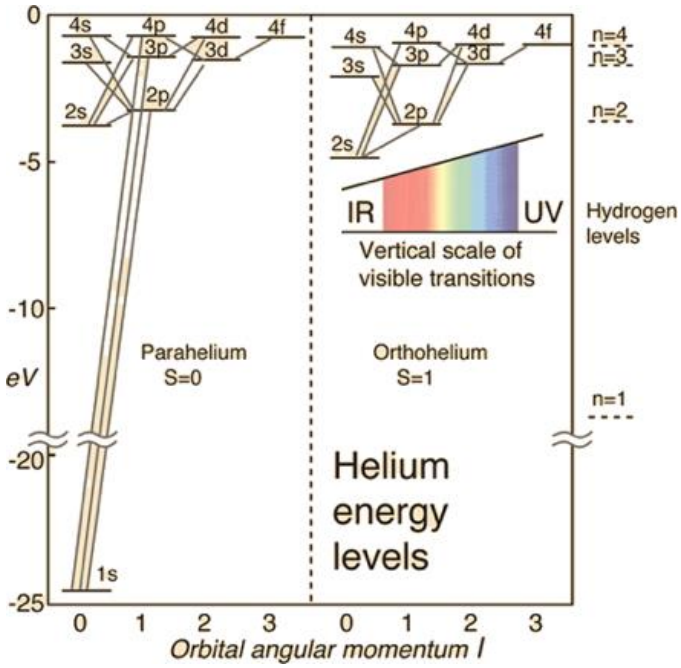


Figure 2. Energy level diagram of Helium. From http://hyperphysics.phy-astr.gsu.edu/hbase/quantum/helium.html

	Initial State	Final State	E_γ (eV)	λ (nm)
Parahelum Transitions	3p	2s	2.472	501.6
	4p	2s	3.127	396.5
	3s	2p	1.702	728.5
	4s	2p	2.455	505.0
	3d	2p	1.856	668.0
	4d	2p	2.519	492.2
Orthohelium Transitions	3p	2s	3.188	388.9
	3s	2p	1.754	706.9
	4s	2p	2.630	471.4
	3d	2p	2.109	587.9
	4d	2p	2.773	447.1

Table 3. The visible allowed transitions between helium states.

Higher-*Z* elements

For other elements, Bohr's model fails in part because we have multiple electrons in our system, causing interactions which shift the allowed energy levels. Even so, we can use it to model spectral lines for one-electron ions (such as He^+, Li^{++}, etc) or atoms with a single valence electron, such as lithium (Li) or sodium (Na). In these cases, the "kernel" electrons "shield" the charge of the nucleus, so the valence electron sees an ***effective nuclear charge*** Z_{eff}. The effective nuclear charge is calculated to be 1.69 for a ground state electron in helium, for example, and 1.28 for the valence electron in lithium.

The Bohr model also works for studying the ***K_α-lines*** of heavier elements, which are the results of transitions from the $n = 2$ to $n = 1$ states, assuming an effective charge of $Z - 1$.

To fully predict the energy levels and spectra of the elements requires a full quantum mechanical treatment. For the purposes of today's lab, it suffices to have a reference for various spectral lines of the elements:

- https://en.wikipedia.org/wiki/Gas-discharge_lamp#Color - Wikipedia's visual table of atomic spectra.
- https://web.archive.org/web/20201103055152/http://chemistry.bd.psu.edu/jircitano/periodic4.html - Click the relevant element to see its visual spectrum.
- http://hyperphysics.phy-astr.gsu.edu/hbase/quantum/atspect.html - Hyperphysics' page on atomic spectra, which contains some details about various atomic spectra of relevance to today's lab.

Some properties of specific elements:

- Hydrogen ($Z = 1$): Characterized by three or four sharp visible lines, as predicted by the Bohr model.
- Helium ($Z = 2$): Helium has a very strong spectral line in the yellow ($\lambda = 587.6$ nm), with several more visible spectral lines in the blue and red regions. Visible wavelengths may include 439 nm, 447 nm, 471 nm, 492 nm, 502 nm, 588 nm, and 667 nm, though other lines are weakly visible.
- Nitrogen ($Z = 7$): Nitrogen shows a large number of distinct bands throughout the visible range,
- Neon ($Z = 10$): Neon is characterized by a large number of red spectral lines, with some noticeable green lines. Of note is the absence of violet lines.
- Sodium ($Z = 11$): Characterized by a single strong spectral line (actually a close pair of lines) called the sodium D-line at $\lambda = 589.3$ nm.
- Argon ($Z = 18$): Argon exhibits a wide range of lines throughout the spectrum.

Creating Emission Spectra

In this lab we will create emission spectra using ***discharge tubes***. Each discharge tube is filled with a gas of a particular element. If we apply a high voltage across the terminals of the discharge tube then an ***electrical discharge*** occurs (sometimes called a ***plasma discharge***) causing a current to flow through the tube. As it travels, this current can transfer energy to electrons in the surrounding gas, exciting the electrons to higher energy states. The excited electrons then emit photons as they relax, resulting in an atomic spectrum.

Note that when we excite the electrons there is nothing preventing us from temporarily ionizing some of the gas. This leads to ***free electrons*** in the gas in addition to excited electrons. These free electrons can have any (positive) energy. Just like an excited electron can fall to a lower energy state, a free electron can get "captured" by an ionized hydrogen, releasing its energy in the form of a photon as it relaxes into a bound state. This produces a continuous spectrum, since there is nothing limiting the initial electron energy. This continuous spectrum will appear as a faint rainbow when we observe the emission spectrum. Our actual spectral lines should appear as brighter lines against the backdrop of this spectrum.

Absorption Spectra

Another way of observing spectra is to study what wavelengths of light get *absorbed* when the light passes through a gas of a particular element. For example, the emission spectrum of the sun is characterized by a hydrogen absorption spectrum - the wavelengths corresponding to the spectral lines of hydrogen are dimmer than expected or missing altogether from the sun's blackbody spectrum. This is because those photons have been absorbed by the electrons in hydrogen, exciting them from a lower to a higher energy state. For example, absorption of photons by elemental hydrogen tells us where clouds of hydrogen exist in the interstellar medium. We can use the spectrum to determine the concentrations of certain elements in stars. We can even use the *Doppler shift* of the spectrum to determine how fast a given star or galaxy is moving away from us! Science is neat…

Fluorescence

In certain cases, an electron in an atom will absorb a high-energy photon and jump to a high energy level. The electron will then de-excite by falling to a lower energy level and emitting another photon. However, it doesn't have to fall *all the way* back down to where it started. It can fall to an intermediate state. In this case the emitted photon has a smaller energy than the absorbed photon. This means a longer-wavelength photon is produced, a phenomenon known as ***fluorescence***. This is the operational principle behind ***CFL*** (compact fluorescent lamp) ***bulbs***. The tube of a CFL contains mercury vapor. The emission spectrum of mercury is characterized by certain strong ultraviolet lines. The surface of the tube is painted with various ***phosphors*** which will absorb the ultraviolet photons and fluoresce in certain visible wavelengths. The "color temperature" of the bulb can be controlled by changing the type of phosphors used in the coating (a 3500K CFL bulb, for example, doesn't operate at 3500K, but produces a spectrum that mimics somewhat a 3500K blackbody spectrum).

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PRE-LAB PREPARATION QUESTIONS

Pre1. Linearizing the Rydberg Equation:

In this lab we will observe the visible wavelengths making up the Balmer series.

a) Use the variables $x \equiv 1/n_i^2$ and $y = 1/\lambda$ to linearize Eq. 1.

b) In the lab we will do a linear fit for $\{x,y\}$ data. What will be the interpretations of the slope m and the y-intercept c?

c) Propagate uncertainties and determine an expression for R and its uncertainty α_R in terms of the fit parameter m and its uncertainty α_m.

Note: This is a straightforward bit and is mainly setup for the next bit.

d) Using your results from (b) and (c), propagate uncertainties and determine an expression for n_f and its uncertainty α_{n_f} in terms of the fit parameters m and c and their uncertainties α_m and α_c.

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EQUIPMENT NOTES

The Spectrometer

[This is repeated from Lab 5] To view the spectrum of the LEDs we will use a ***diffraction grating*** to separate the wavelengths. The grating is mounted on an optical bench in front of the LEDs. At the end of the optical bench is a ***spectrometer scale*** - a mountable ruler with a thin opening in the center (to let light through) and ***slider bars*** to mark position.

Note: To properly view the spectrum, you should get your eye right next to the diffraction grating as shown in Figure 3.

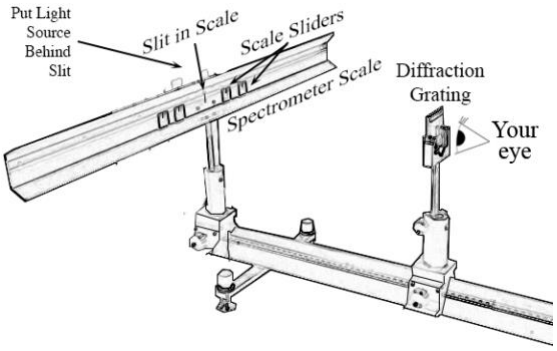


Figure 3: The Spectrometer Setup.

The Discharge Tube Apparatus

To create the spectrum we will use a ***discharge tube***. Our discharge tubes can be inserted into the power supply apparatus as shown. Please have power off whenever you load or unload the discharge tubes.

There are four tubes at each setup: Hydrogen (H, $Z = 1$), Helium (He, $Z = 2$), and some "mystery tubes" labeled either "A", "B", or "C". Tubes A, B, and C can possibly contain the gases Nitrogen ($Z = 7$), Neon ($Z = 10$), and Argon ($Z = 18$) (though not in that order).

The main casing holding the discharge tube can be raised and lowered as shown in Figure 4.

Note that some of the tubes may take between 45 and 90 seconds to fully warm up and turn on. If a tube doesn't turn on immediately please give it some time. If it still has not come on after 90 seconds call the instructor or GSI over.



Figure 4: Left: The discharge tube apparatus. Middle: Loading a tube. Right: Unloading a tube.

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Grating Mirror

The grating mirror, also called a ***reflection grating***, is a piece of spherical mirror with a diffraction grating ruled on part of the reflecting surface which reflects light and forms spectra. Due to the delicate nature of the grating mirror, it is important that you **do not touch the surface of the grating mirror**.

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TIER 1

Experiment 1 – Helium

In this experiment we will see the functioning of the spectrometer and calibrate our scale using some known spectral lines of Helium.

- *This experiment uses the optical bench, discharge power supply, helium discharge tube, diffraction grating, and spectrometer scale.*

1.1 - Calibration

Setup: Load the helium tube into the power supply and set up the spectrometer to observe the emitted light.

Note: You may need to adjust the position and angle of the discharge power supply, the optical bench, and yourself to comfortably and clearly see the spectral lines.

Note: Remember! The main casing holding the discharge tube can be raised and lowered as needed!

Setup: Adjust the distance between the diffraction grating and the spectrometer scale to a comfortable length and record the length.

Note: We will be taking data on spectral lines so you want to be able to easily and accurately find and measure distances. This is a balance between having to look at extreme angles and having the spectrum be spread out enough so that your distance measurements can resolve the differences between the locations of the spectral lines.

Observation: Look through the grating and describe the full pattern you see on the spectrometer scale. Be sure to look on both sides of the central opening and at extreme angles - you should be able to observe more than one set of intensity maxima.

Note: If everything is set up properly you should see the full helium light through the central slit of the spectrometer scale. Off to either side you should see some clear spectral lines, including a very clear, bright yellow line.

Reminder: You need to make sure the spectrometer scale is perpendicular to the optical bench lamp! If it not you will get asymmetries in the spectra on either side of the central slit.

The very bright, clear, yellow line you see is the ***Helium D3 Line***, λ_{D3} = 587.6 nm.[4]

Measure: Use the sliders on the spectrometer scale to find the distance between the yellow spectral line and the central maximum. Each member of your group should do this for the first maxima on either side on the central maximum so that you get a number of independent readings.

Note: You are going to use these independent readings to get an estimate of the uncertainty of your other spectral line measurements!

Measure: Make an estimate of the line width (the thickness of the line) - note that this will probably be too thin to directly measure so just give an educated order-of-magnitude answer. Will the line width significantly contribute to the uncertainty of your measurement?

Reminder: All measured quantities should be recorded with uncertainties!

Note: Make sure the spectrometer scale is properly perpendicular to the axis of the optical bench! Also make sure the diffraction grating isn't wiggling around and is oriented perpendicular to the axis. If your

[4] It was observation of this wavelength in the solar spectrum in 1868 that led to the hypothesis of the element Helium, 27 years before the gas was finally identified.

measurements for distance on either side of the central maximum are consistently different, you may be at an angle.

Once you've calibrated, be careful not to move the bench stands containing the spectrometer scale or the diffraction grating. If things get moved, you may need to re-calibrate!

1.2 - Helium Spectrum!

In the Theory and Background section the predicted spectral lines of helium were discussed. Let's test this!

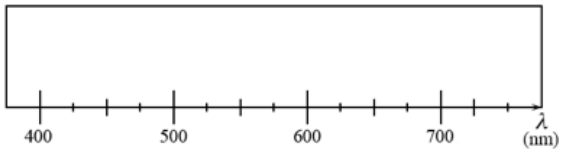
Observation: Sketch and label the observed first order spectral lines on either side of the central maximum.

Note: You most probably won't be able to find all of the predicted lines. That's okay! Some lines are very faint. Each tube is also somewhat different - sometimes you can see a lot of clear spectral lines and other times you will have difficulty finding any but the brightest.

Data Collection: Make a data table of spectral line (based on label) vs. distance from the central maximum (with uncertainty).

Analysis: Determine (without uncertainty for now) the wavelength for each line. For each wavelength, describe (according to you) the observed color (*e.g. yellow/orangish, teal, deep blue, etc.*).

Visualization: Make a "spectral line plot" of your observed spectral lines by wavelength. (Basically, draw where the spectral lines appear on a version of the "graph" to the right).



Analysis: Compare your results with the predicted (or the listed) wavelengths. Do they roughly agree?

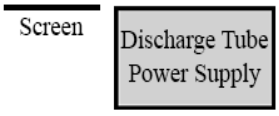
Experimental Design: Based on your observations, come up with a good "fingerprint" for helium. That is, identify some clear features that would let you look at a spectrum and quickly identify it as helium.

At your table is a spherical grating mirror (600 lines/mm) and a screen in a free stand. This will let us properly project the spectrum onto a screen, as shown in the diagram to the right.



Warning: Do not touch the surface of the grating mirror.

Setup: Arrange the power supply, grating mirror, and screen so you form an in-focus image of the spectrum on the screen.



Observation: Trace the locations of each identifiable spectral line to get another version of your sketch from earlier. Describe any significant discrepancies between your two observations. In particular, look for any new spectral lines due to ultraviolet light fluorescing on the paper! If you identify any new lines, determine their wavelengths.

Experiment 1 Checkpoint
Call the instructor over to discuss Experiment 1 at this point before moving on.

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Experiment 2 – The Bohring World of Hydrogen

In this experiment we will observe the Balmer series of hydrogen!

- *This experiment uses the optical bench, discharge power supply, hydrogen discharge tube, diffraction grating, and spectrometer scale.*

2.1 - The Balmer Series

Setup: Load the hydrogen tube into the power supply.

Observation: Describe, sketch, and label the observed first order spectral lines on either side of the central maximum.

Note: You are looking for bright distinct bands. Unfortunately, the hydrogen tubes we use also produce many free/ionized electrons. This contributes to an overall faint continuous spectrum in the background.

Data Collection: Make a data table of spectral line (based on label) vs. distance from the central maximum (with uncertainty).

Note: You should be able to find at least three spectral lines. The fourth is iffy and difficult to see. Don't worry if you only can collect the three data points.

Analysis: Determine (without uncertainty for now) the wavelength for each line. For each wavelength, describe (according to you) the observed color (*e.g. yellow/orangish, teal, deep blue, etc.*).

Analysis: Compare with the predicted wavelengths for the Balmer series from the Rydberg formula Eq. 1. Do they roughly agree? Label each spectral line on your sketch with the appropriate energy level transitions. [*e.g. 4→2 for the line corresponding to the wavelength of an n=4 to and n=2 state*.]

Note: Some lab groups in the past have shown systematic error in their measurements. This could be from an incorrect calibration or changing the spectrometer-to-grating distance. If your numbers are off, check to see if they are consistently off ("my numbers were all low by about 15 nm" or "my numbers were all off by a factor of 1.1"). Even if you don't match exactly you should be able to properly identify which line goes with which transition.

Experimental Design: Based on your observations, come up with a good "fingerprint" for hydrogen. That is, identify some clear features that would let you look at a spectrum and quickly identify it as hydrogen.

Data Collection: Using the grating mirror as we did for helium, project the hydrogen spectrum onto a screen and sketch the locations of each identifiable spectral line. In particular, look for any new spectral lines due to ultraviolet light fluorescing on the paper. *If* you are able to identify a new spectral line, take data to determine the wavelength.

Experiment 2 Checkpoint
Call the instructor over to discuss Experiment 1 at this point before moving on.

- *At this point you have taken all the relevant data for this experiment. The rest of this experiment involves analyzing the data and will take some time*.

2.2 - The Bohr Model for Hydrogen

a) Find the standard deviation and standard error for your multiple independent measurements for the *position* of the Helium D3 line from Experiment 1.1.

Note: The standard deviation will act as one of the main source of uncertainties for position measurements of other spectral lines using our spectrometer scale. The standard error will act as the uncertainty for the position of the helium D3 spectral line.

The Bohr model predicts a linear relationship between $1/\lambda$ and $1/n_i^2$ for the Balmer series of hydrogen, with a predicted value for the *y*-intercept and an expression for the slope in terms of the Rydberg constant *R*. Recall that we determined the appropriate n_i values for our data points in Part 2.1(d).

b) Make a scatter plot (with error bars) of $1/\lambda$-vs-$1/n_i^2$.

c) Perform a regression to the linear function you found in Question Pre1(a) to find a best-fit value for the slope (with uncertainties).

Note: After you propagate uncertainties, you may find that you need to do a weighted fit rather than a simple fit.

Note: You only have three or four data points, so it's not going to be a great fit necessarily, but it'll do for us!

Note: We can take the predicted y-intercept as a given parameter based on the assumption $n_f = 2$ and not letting it be a free parameter.

d) Based on your fit values, determine the Rydberg constant (with uncertainty). Compare to the given value.

2.3 - Bohr Fails for Helium [Optional]

Our visible transitions of helium all ended in one of the $n = 2$ levels, so maybe we can make Bohr work by assuming the nucleus has an *effective* charge Z_{eff} due to one of the two electrons partially screening the true nuclear charge of $Z = 2$.

Find a reasonable assignment of n_i values to your visible spectral lines of helium and fit to the Bohr model using the linear relationship between $1/\lambda$ and $1/n_i^2$ from earlier. Find the best-fit slope for this fit. Using the known Rydberg constant, use your answer to determine Z_{eff}.

Perform a statistical test to show that this is a particularly poor model that doesn't fit the data well at all.

Note: To assign the n_i you should start your longest wavelength at $n_i = 3$ and each successive shorter wavelength should be the next n_i. As a challenge, you could of course try different assignments of integers n_i to the various spectral lines. The point is, helium's spectrum does not follow the Bohr model so no assignment of integers is really expected to work all that well…

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TIER 2 – A SPECTRUM OF SPECTRA!

Experiment 3 – Mystery Tubes!

At each setup, you have two "mystery tubes". You all have a *B* and then either an *A* or a *C*. The possibilities for these gases are Nitrogen ($Z = 7$), Neon ($Z = 10$), and Argon ($Z = 18$). For each of your two mystery tubes, determine what the unknown gas is! Try to pick out one or two of the most prominent spectral lines and measure their wavelength. Research to verify whether the element you determined has your measured wavelengths as part of their emission spectra.

*Note: Your comparison with reported spectra **shouldn't** just be "they agree" or "Gas C's spectral lines roughly align with this picture's and so we posit that Gas C is Nitrogren."*

Note: Be sure to make detailed descriptions of the spectral lines and discuss what features let you determine the element. That is, describe the "fingerprint" you used to identify the element.

An ideal analysis would include explicitly comparing some of the wavelengths of measured spectral lines to known lines in the gas' spectrum. You should also make some detailed qualitative observations that support your identifications (noting gaps, broad continua, etc.) and should perform some statistical tests to back up your claim!

Experiment 3 Checkpoint
Call the instructor over to discuss Experiment 3 at this point before moving on.

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Experiment 4 – Light Bulbs

a) Set up your spectrometer to view the light from the optical bench lamp at your table. This contains an ***incandescent bulb***. Observe and describe the spectrum of the incandescent bulb. Use this spectrum to determine (your) visible wavelength range! Make note of the wavelengths your eye seems to be most sensitive to as well.

Note: This will obviously involve judgement calls on the bounds of the spectrum you can actually see.

b) At a side table is a spectrometer setup with an optical bench lamp containing a ***compact fluorescent lightbulb*** (CFL), which are discharge tubes containing mercury vapor and coated with fluorescent paint. Observe and describe the spectrum of the CFL, noting the colors, wavelengths and rough line widths of each spectral line. Do any of these coincide with the spectral lines of mercury?

Note: You can't really use this data to determine which spectral lines of mercury are actually causing the fluorescence of the paint, but you can put some bounds on which lines are responsible (it turns out that the 185 nm and 254 nm are primarily culprits).

Experiment 4 Checkpoint
Call the instructor over to discuss Experiment 4 at this point before moving on.

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FOR FUN

From the column "The Wit Parade" in the magazine "The American Weekly" in September 1956:

A friend was visiting in the home of Nobel Prize winner Niels Bohr, the famous atom scientist. As they were talking, the friend kept glancing at a horseshoe hanging over the door. Finally, unable to contain his curiosity any longer, he demanded: "Niels, it can't possibly be that you, a brilliant scientist, believe that foolish horseshoe superstition!?!" "Of course not," replied the scientist. "But I understand it's lucky whether you believe in it or not."

Another version of this I really like, due to the whole Bohr model thing, has Bohr responding “I don’t, but it seems to work anyway”.

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